

## Module 5

# Valuing Bonds and Stocks

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FINE 3010, Financial Management  
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# From Module 4 to Module 5

- **Module 4**, the time value of money, the present value of future cash flows

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- **Module 4**, the time value of money, the present value of future cash flows
- **Module 5**, the same time-value-of-money principles value bonds and stocks
  - Bond, the present value of its coupons and face value
  - Stock, the present value of its expected dividends

# Every Financial Asset Is Debt or Equity

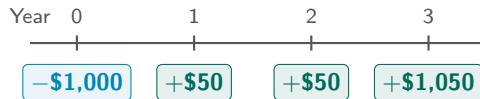
- **Financial asset**, value from a contractual right or ownership claim



# Debt's Cash Flow Is Fixed, a Stock's Is Not

## Debt

\$1,000, 3 years, 5% interest

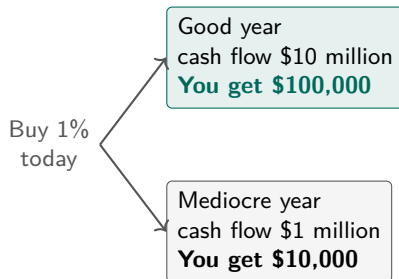


Every payment known in advance

## Stock

1% of the company

Next year



Value of claim depends on how the company does

# Common Types of Corporate Financing

**Debt**, lending to the company

Main forms, bonds and loans

Holder gets a fixed promised amount

“Senior”, paid back before equity in  
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No voting rights or control

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**Equity**, stock ownership in the company

Holder gets dividends

“Residual claimant”, paid whatever is left after all promised payments, no guaranteed payout

Voting rights

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## Module 5, Part 1

# Bonds

# Bonds, Definitions

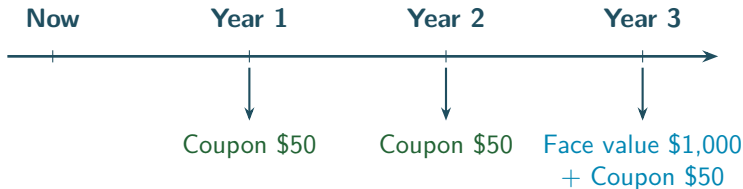
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- **Bond**, a security that obligates the issuer (the borrower) to make specified payments to the investor (the lender, or “bondholder”)
- Typically periodic payments (the “coupon”), then a “face value” at the end
- E.g., a three-year bond with an annual coupon of \$50 and a face value of \$1,000, the borrower promises to pay,



# Bonds, Terminology

- **Face value**, the final lump-sum payment at the “maturity” of the bond, usually \$1,000 per bond in the U.S.



Imaged by Heritage Auctions, HA.com

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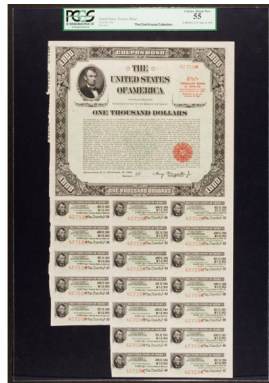


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- **Coupon rate**,

$$\text{Coupon rate} = \frac{\text{Annual coupon}}{\text{Face value}}$$



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  - E.g., **Module 4**'s Puerto Rico bond, the coupon was 0 but the interest rate was 5.15%

# Example, Puerto Rico Borrows

Recall from **Module 4**

- In 2007, Puerto Rico sold bonds that promised to pay \$1,000 after 47 years (no periodic “coupon” payments)
- These bonds were sold at \$94.40 each
- What was the interest rate paid by Puerto Rico?



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Recall from **Module 4**



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- Solution,
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  - -94.40 **PV**
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$$\text{Value} = \text{PV} = \frac{\text{Coupon}_1}{(1+r)^1} + \frac{\text{Coupon}_2}{(1+r)^2} + \dots + \frac{\text{Coupon}_T + \text{Face value}}{(1+r)^T}$$

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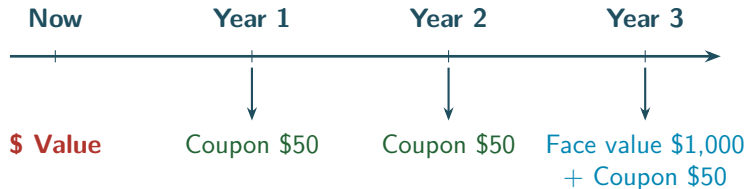
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- Investors choose the opportunity cost, “what could I get investing in something else with a similar risk”, as the required rate of return  $r$

## Example, Bond Pricing

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$$\text{Value} = \frac{\$50}{(1 + 0.04)^1} + \frac{\$50}{(1 + 0.04)^2} + \frac{\$50 + \$1,000}{(1 + 0.04)^3} = \$1,027.75$$

## Example, Bond Pricing with a Financial Calculator

- The same bond, an annual coupon of \$50, a face value of \$1,000, three years to maturity, and a required rate of return of 4%
- Use a financial calculator to value the bond

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  - 4 **I/Y**
  - **CPT**, then **PV**

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- The output is  $-\$1,027.75$  (why the minus sign? recall how financial calculators get to the answer in [Module 4](#))

# Valuing Bonds with a Financial Calculator

- The financial calculator keys map onto a bond,
  - **N**, the number of periods (years) until maturity
  - **PV**, the value of the bond (the present value of the coupons and the face value)
  - **PMT**, the coupons (in dollars)
  - **FV**, the face value (the lump sum paid at the end)
  - **I/Y**, the discount rate (the required rate of return  $r$ )

# Recap of Part 1

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- In return, the companies make specified payments of coupons and face value to the investors
- The value of a bond is the discounted payments, at the required rate of return

## Module 5, Part 2

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- **Yield**, the discount rate  $y$  for which the present value of the bond's payments (coupons and face value) equals its price
- Given the price of a bond (PV), its maturity  $T$ , and its coupons, find the yield by solving for  $y$ ,

$$\text{Price} = \frac{\text{Coupon}}{(1+y)^1} + \frac{\text{Coupon}}{(1+y)^2} + \cdots + \frac{\text{Coupon} + \text{Face value}}{(1+y)^T}$$

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Recall from **Module 4**

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- **Here we are calculating the yield!**



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## Example, Bond Yields

- Solving for the yield analytically is hard, so use the financial calculator
- A bond's annual coupon rate is 5%. Its face value is \$1,000. Its maturity is 3 years. Its current market price is \$1,027.75. What is its yield?

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  - **CPT**, then **I/Y**
- The yield is 4%

# Required Rate of Return vs. Yield

## Required rate of return $r$

- The discount rate for computing the bond's value
- Chosen by investors as the opportunity cost, the return on alternatives with a similar risk
- Input  $r$ , output the value  
(**CPT** **PV**)

## Yield $y$

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- The discount rate for which the present value of the payments equals the price
- Implied by the market price
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# Required Rate of Return vs. Yield

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## Yield $y$

- The discount rate for which the present value of the payments equals the price
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- Input the price, output the yield  
(**CPT** **I/Y**)
- **Objective**

## Practice Problem

- The table shows data for three bonds. Each bond's coupon is zero. Each bond's face value is \$1,000

| Bond | Price | Maturity (years) | Yield |
|------|-------|------------------|-------|
| A    | \$310 | 25               | ?     |
| B    | \$310 | ?                | 9%    |
| C    | ?     | 11               | 8%    |

- What is the yield of bond A?
- What is the maturity of bond B?
- What is the price of bond C?

## Practice Problem

- Solution, on the financial calculator (the highlighted key is the one solved for),

|           | <b>N</b> | <b>I/Y</b> | <b>PV</b> | <b>PMT</b> | <b>FV</b> |
|-----------|----------|------------|-----------|------------|-----------|
| a. Bond A | 25       | ?          | −310      | 0          | 1,000     |
| b. Bond B | ?        | 9          | −310      | 0          | 1,000     |
| c. Bond C | 11       | 8          | ?         | 0          | −1,000    |

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|           | N  | I/Y   | PV   | PMT | FV     |
|-----------|----|-------|------|-----|--------|
| a. Bond A | 25 | 4.796 | -310 | 0   | 1,000  |
| b. Bond B | ?  | 9     | -310 | 0   | 1,000  |
| c. Bond C | 11 | 8     | ?    | 0   | -1,000 |

- a. The yield of bond A is 4.796%

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- a. The yield of bond A is 4.796%
- b. The maturity of bond B is 13.59 years

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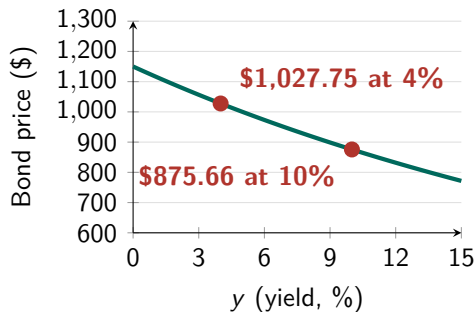
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- a. The yield of bond A is 4.796%
- b. The maturity of bond B is 13.59 years
- c. The price of bond C is \$428.88

# Bond Prices and Yields Move in Opposite Directions

- Because the bond price is,

$$\text{Price} = \frac{\text{Coupon}}{(1+y)^1} + \frac{\text{Coupon}}{(1+y)^2} + \dots + \frac{\text{Coupon} + \text{Face value}}{(1+y)^T}$$

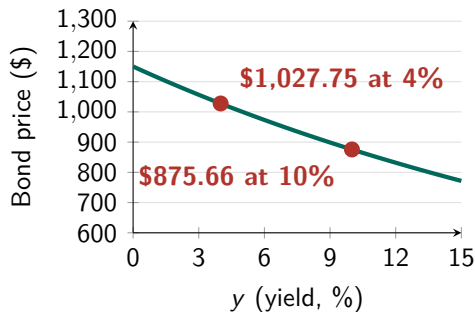


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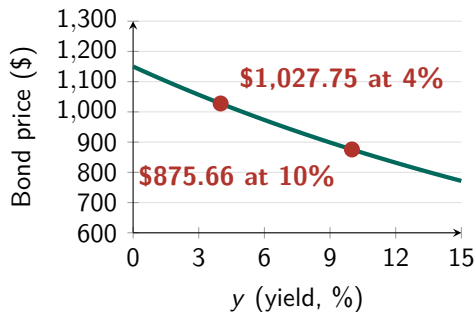


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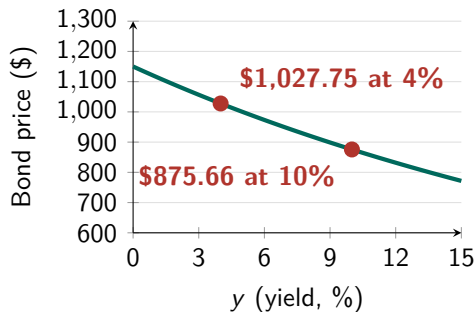


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1. Market (risk-free) interest rates (time value of money)
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2. Risk of default (compensation for taking risks)
  - Higher risk of default, lower price, higher yield

# Default Risk

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- **Default premium**, the additional yield on a bond that investors require for bearing default risk

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- **High-yield bonds** ("junk" bonds), rated Ba or BB and lower

## Default Risk, Credit Ratings

| Moody's                       | S&P     | Default within 10 yr |
|-------------------------------|---------|----------------------|
| <b>Investment-grade bonds</b> |         |                      |
| Aaa                           | AAA     | 0.1%                 |
| Aa                            | AA      | 0.7%                 |
| A                             | A       | 2.2%                 |
| Baa                           | BBB     | 3.5%                 |
| <b>High-yield bonds</b>       |         |                      |
| Ba                            | BB      | 15.7%                |
| B                             | B       | 35.5%                |
| Caa, Ca                       | CCC, CC | 48.9%                |
| C                             | C       | n/a                  |

Source, Brealey, Myers, and Marcus, Table 6.2

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- Yields tend to be higher (prices lower) for lower-rated bonds

# Treasury Yields at 20-Year Highs

- **Treasury yield**, the yield on bonds issued by the U.S. government (“Treasury bonds”)



PBS NewsHour, August 20, 2026 (7:14), “Why Treasury yields are at 20-year highs”

## Recap of Part 2

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- Bond prices and yields move in opposite directions
- Prices fall and yields rise with market interest rates and with default risk, lower-rated bonds carry higher yields

## Module 5, Part 3

# Stocks

# Goals

- **Part 1**, bonds
  - Differences between debt and equity
  - Valuation
- **Part 2**, bond yields
  - Yields
  - Interest rates
  - Default risk
- **Part 3**, stocks
  - Valuation, dividend discount model
  - Valuation, comparables

# Valuing Stocks

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  1. Dividend discount model
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- More in “FINE 4100 Advanced Financial Management” and “FINE 4110 Investments in Equities”

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- **Dividend yield**,  $\frac{\text{Dividend}}{\text{Stock price}}$
- **Earnings per share (EPS)**,  $\frac{\text{Net income}}{\text{Number of shares outstanding}}$
- **P/E ratio**,  $\frac{\text{Stock price}}{\text{Earnings per share}}$

# Valuing Stocks, Dividend Discount Model

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- **Required return on equity** ( $r_e$ ), the discount rate, set by the opportunity cost of investing in something with similar risk

# Valuing Stocks, Dividend Discount Model

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- What price should you pay for the stock today ( $P_0$ )?
- Stock value, the present value of the cash flows you get from owning it,

$$P_0 = \frac{\text{Div}_1}{(1 + r_e)^1} + \frac{\text{Div}_2}{(1 + r_e)^2} + \frac{\text{Div}_3}{(1 + r_e)^3} + \cdots + \frac{\text{Div}_\infty}{(1 + r_e)^\infty} = \text{PV}(\text{Dividends})$$

- **Required return on equity** ( $r_e$ ), the discount rate, set by the opportunity cost of investing in something with similar risk
- $\text{Div}_1, \text{Div}_2, \dots$  are *expected* dividends

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- The second formula is often called the “**Gordon growth model**”

## Example, Gordon Growth Model

- 1. A firm will pay a \$3.00 dividend forever (the first dividend exactly a year from now). Its required return on equity is 12%. What will investors pay for the stock?
- 2. What if the firm's stock pays a \$3.00 dividend a year from now, and then the dividend grows at 8% forever?

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- That is, dividend yield plus expected capital appreciation (in %)

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  - If the firm is only temporarily not paying dividends, the dividend discount model still applies
  - But future dividends are then even more uncertain!

# Valuing Dividends with Non-Constant Growth

- Estimate the dividends year by year over a horizon of  $H$  years, then assume constant growth after year  $H$ ,

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- More in “FINE 4100 Advanced Financial Management” and “FINE 4110 Investments in Equities”

## Example, Valuing Non-Constant Growth

- A stock's expected dividends are in the table. Its required return on equity is 8.5%. Its dividends grow at 6% a year beginning in year 6. What is the value of the stock?

| Year     | 1      | 2      | 3      | 4      | 5      |
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$$PV_{\text{Stock}} = 10.23 + 89.11 = \$99.33$$

## Practice Problem

- Consider the following three stocks,
  - a. Stock A is expected to pay a dividend of \$10 a share forever
  - b. Stock B is expected to pay a dividend of \$5 next year. Thereafter, dividend growth is expected to be 4% a year forever
  - c. Stock C is expected to pay a dividend of \$5 next year. Thereafter, dividend growth is expected to be 20% a year for 5 years (years 2 through 6) and zero thereafter
- Each stock's required return on equity is 10%. Which stock is the most valuable? What if the required return on equity is 7%?

# Practice Problem

- Solution, at a 10% required return on equity,

10% capitalization rate:

$$P_A = \text{DIV}_1 / r = \$10 / 0.1 = \$100$$

$$P_B = \text{DIV}_1 / (r - g) = \$5 / (0.1 - 0.04) = \$83.33$$

$$P_C = \frac{\text{DIV}_1}{1+r} + \frac{\text{DIV}_1 \times (1+g)}{(1+r)^2} + \frac{\text{DIV}_1 \times (1+g)^2}{(1+r)^3} + \frac{\text{DIV}_1 \times (1+g)^3}{(1+r)^4} + \frac{\text{DIV}_1 \times (1+g)^4}{(1+r)^5} +$$

$$\frac{\text{DIV}_1 \times (1+g)^5}{(1+r)^6} + \frac{(\text{DIV}_1 \times [1+g]^5 \times [1+g_2]) / r}{(1+r)^6}$$

$$P_C = \$5 / 1.1 + (\$5 \times 1.2) / 1.1^2 + (\$5 \times 1.2^2) / 1.1^3 + (\$5 \times 1.2^3) / 1.1^4 \\ + (\$5 \times 1.2^4) / 1.1^5 + (\$5 \times 1.2^5) / 1.1^6 + ([\$5 \times 1.2^5 \times \{1 + 0\}] / 0.1) / 1.1^6$$

$$P_C = \$104.51$$

At a 10% capitalization rate, Stock C has the largest present value.

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At a 10% capitalization rate, Stock C has the largest present value.

- At a 7% required return on equity,  $P_A = \$142.86$ ,  $P_B = \$166.67$ , and  $P_C = \$156.50$ , so Stock B is the most valuable

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- E.g., a 2,000 sqft house in New Orleans sells for \$400,000. What is your estimate for a neighboring house of 2,500 sqft?

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- Underlying idea, we can “scale” prices!

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- 4. Back out Company X’s implied value (the numerator) as the average multiple times Company X’s denominator

$$P_X = (\text{Average P/E of comparables}) \times \text{EPS}_X$$

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- Solution, multiply Herman Miller's EPS by the P/E of comparable firms,  
$$P = (\text{Average P/E of comparables}) \times \text{EPS} = 21.3 \times 1.38 = \$29.39$$

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- The question asks for the share price, not the market capitalization,

$$P = \frac{\text{Market cap}}{\text{Number of shares outstanding}} = \frac{36.8 \text{ billion}}{1.2 \text{ billion}} = \$30.67$$

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- Assumes the comparables are priced correctly
  - Cannot tell if an entire industry is overvalued or undervalued
  - E.g., no help in valuing tech firms during the late-1990s “dot-com bubble”

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# Thank you!

Please let me know if you have any questions.